

ANTHONY D. CARPENETTI

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DESIGN ENGINEER

PREFERRED NAME: TONY

Mechanical Design Engineer with 12+ years of experience delivering solutions across aerospace and other industries, including 5+ years with NASA's Space Launch System (SLS) Artemis Program as a vehicle integration engineer with the EV32, Structural Design, Development, and Analysis Branch. Currently analyzing the tolerance alignment for the entire SLS hardware envelope which directly impacts the accuracy and safety of the design. Proven track record combining technical expertise with strong communication skills to complete individual and team tasks while supporting NASA's mission.

Core competencies: Aerospace Engineering; Design and Development; Technical Writing and Presentation; Project Management; CAD Modeling and Simulation; Computer Programming; Communication; Critical Thinking; Public Speaking

CAREER HIGHLIGHTS

- Performed design analysis on Artemis II flight hardware which contributed to successful 99.92% Insertion Accuracy **APR. 2026**
- Received the August 2022 Outstanding Task Award from Amentum for successful accelerated delivery of Mechanical Interface Control Drawings (MICDs) for NASA's Artemis II Block 1B Exploration Upper Stage (EUS) Forward and Aft Umbilical Interfaces

PROFESSIONAL EXPERIENCE

NASA, Marshall Space Flight Center (MSFC) Huntsville, AL (40+ hrs./week) **JUN. 2026 – PRESENT**

Aerospace Engineer, AST, Aerospace Flight Systems

SLS Artemis Program – [EV32, Structural Design, Development, and Analysis]

FEB. 2021 – PRESENT

Provide engineering design support for hardware design cycles and flight/ground interfaces, integrating large assemblies and subassemblies while working individually and with a team of engineers to develop aerospace vehicles, components, and systems. Produce formal reports, technical presentations and reviews of technical data; develop CAD models and drawings.

- Accomplished successful delivery of 30+ design reviews for Block 1, Block 1B, and Block 2 flight configuration projects by documenting and correcting booster nozzle rotational errors, adjusting reference designator misalignments, closing broken model geometries, developing vehicle integration documents, and mitigating various design problems and inconsistent data
- Evaluated a design discrepancy between reported and actual tolerances on the Block 1B EUS Procurement Control Drawing by interpreting the local vs cumulative tolerance data and uncovering a reporting issue in the customer's reporting process
- Led a month-long review cycle for the Block 1B static tolerance envelope design, iterating with customers to refine the data, perform overlay comparisons, and track comments to ensure 100% design accuracy to NASA requirements standards
- Improved the data validation and acceptance process as measured by 50% reduction in required time and significant removal of human error by creating an automated Excel Query for pulling new data and populating the associated tolerance stacks
- Redesigned an Engineering Parts List (EPL) template with improved formatting and increased visibility, accuracy, and usability

Human Landing System (HLS) – [EV73, Systems Analysis and Human Systems Integration]

APR. 2026 – PRESENT

Coordinate the design, planning, and construction of physical mockups for HLS flammability testing efforts. Interface with a multidisciplinary team of engineers, participating in reviews, meetings, working groups, and task teams.

- Developed the test frame structure to mount and secure HLS components within a pressure chamber for flammability testing
- Achieved same-day turnaround for a test chamber CAD mockup after the customer provided a late update to the dimensions

BEVILACQUA RESEARCH CORPORATION (BRC) Huntsville, AL

(40+ hrs./week) **FEB. 2021 – JUN. 2026**

Mechanical Design Engineer on NASA's ESSCA Contract

Performed same duties as listed above before converting from contractor to NASA civil servant in June 2026.

REGAL REXNORD Tipp City, OH

(40 hrs./week) **NOV. 2018 – FEB. 2021**

Mechanical Design Engineer

Own the mechanical design of electric motors for application in pool pumps and commercial ventilation systems. Manage through prototype concept, CAD modeling simulation, benchmark testing, supplier quoting, manufacturing, and customer sign-off.

- Led the design and production of a motor frame retrofit which avoided new tooling and reduced project costs by \$20,000; Modified an existing end frame to fit a larger bearing to meet the need of a sturdier motor at the same size and price bracket
- Drove the innovation of a novel hall sensor mounting solution to overcome space constraints within a motor housing
- Supported \$37.61M cost-out savings with a team using Value Analysis and Value Engineering (VAVE) and 80/20 methodology

HITACHI ASTEMO Mt. Sterling, OH

(40 hrs./week) **SEPT. 2017 – NOV. 2018**

Project Engineer

Lead lifecycle management, quality, and delivery of new product releases as a tier-1 OEM automotive HVAC assembly supplier to Honda and Ford. Implement the ramp-up and fulfillment of model changes while maintaining up-time of the manufacturing line.

- Reduced evaporator core assembly time by 20% via the implementation of a robotic assembly arm. Programmed the Human Machine Interface (HMI) to reduce the initial scrap rate and achieve 97% good part throughput within 3 months
- Designed assembly support fixtures, installed and repaired production line equipment, and improved flow paths and ergonomics

BWI GROUP Dayton, OH | **BASIC SYSTEMS, INC** Cambridge, OH
Mechanical Engineering Intern

(Seasonal 40 hrs./week) **MAY 2014 – MAY 2017**

Research, develop, and design automotive suspension components and systems. Produce technical specifications, request quotes, and oversee implementation of automotive manufacturing equipment. Perform oil and gas fluid dynamics calculations, produce compressor station blueprints, and confirm with on-site measurements.

- Analyzed 1 million+ cycles of damping test data to confirm the reliability of magnetorheological fluid suspension systems
- Diagnosed a \$9,000 manufacturing cross-contamination issue and presented an intermediate parts-washing process solution
- Fostered international and cross-cultural teamwork and communication with foreign production facilities and coworkers
- Produced technical specifications, Process Failure Mode and Effects Analyses (PFMEAs), Process Flow Diagrams (PFDs), Process Control Plans (PCPs), & quotes. Honed CAD drafting skills while adhering to ASME Y14.5 standards and procedures

SKILLS AND TOOLS

Design Tools: CREO, SolidWorks, ANSYS FEA, AutoCAD, LabVIEW, MATLAB & Simulink, Windchill PLM, Mathcad, CNC, 3D Printing

Miscellaneous: Microsoft Office (Word, Excel, PowerPoint, Project, Outlook, OneNote, Teams) Adobe Acrobat Shared Reviews

Programming & Web Dev Software: Git, GitHub, HTML, CSS, JavaScript, Typescript, Python, Visual Basic, SQL, C++, C, AI

Creative: Affinity (Designer, Photo, Publisher), Digital Audio Workstation (Ableton Live, FL Studio), Adobe, Canva

EDUCATION

WRIGHT STATE UNIVERSITY Dayton, OH 45435

AUG. 2013 – MAY 2017

B.S. Mechanical Engineering | *Cum Laude - Honors Diploma, GPA: 3.53*

COMMUNITY LEADERSHIP

STEM Advocate – Delivered Mechanical Engineering career presentation to 100s of students at Alma Mater HS Career Expo. Member of Amentum Careers Network (ACN), ASED Mentor Peers Program (MPP), Marshall Early Career Organization (MECO)

Local Church Leader – Mentor young adults, prepare weekly study content, develop digital content, play on the worship team

Technology Consultant – Serve as an informal technology advisor for an extended network of family and friends: Diagnose hardware, software, and connectivity issues while translating complex technical solutions into easily digestible language. Apply creative and technical skills across personal and professional projects in web development, graphic design, and digital media – examples available at <https://www.tonycarpenetti.com>. My LinkedIn page --> <https://www.linkedin.com/in/tonycarpenetti>

OTHER ACHIEVEMENTS

Certificate of Completion for the following:

- Tolerance Stack Analysis for both coordinate and geometric tolerances **SEPT. 2025**
- Jacobs (now Amentum) Technical Excellence Development Program (ATED) **OCT. 2023 – JUNE 2024**
- Advanced Concepts of GD&T based on ASME Y14.5-2009 Standard **JUNE 2023**
- Fundamentals of GD&T based on ASME Y14.5-2009 Standard **AUG. 2022**
- Fundamentals of Geometric Dimensioning and Tolerancing (GD&T) **MAR. 2021**